# DemoBank Automated QA Audit Report

## 1. Executive Summary

This report documents the results of the specification-driven DemoBank QA execution pipeline.

- Total test scenarios: **42**
- Passed: **38**
- Failed: **4**
- Unverified: **0**
- Pass rate: **90.48%**
- Confirmed application defects: **4**

## 2. Test Execution and Coverage

The project uses specification-derived requirements and test scenarios with browser automation, application API assertions, and PostgreSQL verification.

| Metric | Result |
|---|---|
| Test scenarios evaluated | 42 |
| Passed | 38 |
| Failed | 4 |
| Unverified | 0 |
| Valid screenshot references | 41 |

The full machine-readable results are available in qa_report.json.

## 3. Confirmed Defects

| Defect ID | Test ID | Finding | Status |
|---|---|---|---|
| DB-QA-003 | TC-018 | Zero-dollar transfer accepted | FAIL |

| Defect ID | Test ID | Finding | Status |
|---|---|---|---|
| DB-QA-004 | TC-019 | Negative transfer accepted | FAIL |
| DB-QA-002 | TC-022 | Insufficient-funds transfer accepted | FAIL |
| DB-QA-001 | TC-031 | Inclusive end-date filtering incorrect | FAIL |

### Observed Failure Details

**TC-018**

Amount='0'; HTTP 201; balances unchanged=True; no success message=False; response={"transaction":{"id":"25aa01e5-7598-4df6-a7a6-7e7ccce10f33","type":"transfer","status":"completed","amount":"0.00","memo":null,"occurred_at":"2026-10-20T06:21:20.466068Z","source_account":{"id":"00000000-0000-0000-0000-000000000101","name":"Everyday Checking","masked_number":"•••• 1042"},"destinatio

**TC-019**

Amount='-10'; HTTP 201; balances unchanged=False; no success message=False; response={"transaction":{"id":"4cf0bfdb-1fc7-45c4-911b-86df0480b984","type":"transfer","status":"completed","amount":"-10.00","memo":null,"occurred_at":"2026-10-20T06:21:20.682992Z","source_account":{"id":"00000000-0000-0000-0000-000000000101","name":"Everyday Checking","masked_number":"•••• 1042"},"destinat

**TC-022**

DB-QA-002: Insufficient-funds transfer was accepted with HTTP 201. Checking balance changed from 4369.75 to -100.00. Expected HTTP 409 and unchanged balances.

**TC-031**

BUG DB-QA-001: Annual bonus (+$950.00) occurred on June 15, 2025, but was excluded when the start and end dates were both June 15, 2025. The PRD requires inclusive date boundaries in Pacific Time.

See DEFECTS.md for reproduction steps, expected behavior, and actual behavior.

## 4. Evidence Audit

- Main report: **41 valid screenshot references**.

- TC-022 has recorded HTTP response and balance-change evidence in the test result, but no screenshot attachment.
- The final-transfer report contains five passing saved results with five valid screenshot references.
- Evidence paths are relative to the agent directory.
- Evidence files are stored under evidence/.

## 5. API and Token Cost Audit

| Cost category | Estimate | Basis |
|---|---|---|
| Paid LLM API calls during deterministic QA execution | 0 | No paid LLM API required by this execution path |
| Paid LLM input/output tokens during QA execution | 0 | No paid LLM API required by this execution path |
| Estimated paid LLM API cost | $0.00 | Deterministic execution only |
| Browser, database, and local compute | Not measured | Local environment |
| Development assistance costs | Not measured | Excluded from execution estimate |

The $0.00 estimate applies only to paid LLM API usage in the deterministic QA execution path. It is not a claim that the entire project had zero development or operating cost.

## 6. Execution Safety and Limitations

- Transfer tests can modify application database balances.
- A controlled successful-transfer run was performed and its results were saved separately.
- The test database was restored after the controlled run.
- The pipeline is specification-driven and deterministic; it does not claim autonomous LLM planning during execution.
- TC-022 does not have a screenshot attachment.
- Full reproduction requires the DemoBank application, PostgreSQL, Python dependencies, and browser tooling.

# 7. Audit Artifacts

- Requirements
- Test plan
- Complete QA report
- Final-transfer results
- Confirmed defects
- Execution instructions

# 8. Conclusion

All 42 planned scenarios received a recorded outcome. The execution identified four application defects and preserved supporting results and available evidence.